Barriers and Facilitators to Adopting Smart Home Fall Prevention Technology: A Qualitative Focus Group Study

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ABSTRACT

Background: Falls are a leading cause of injury and loss of independence among older adults. While smart home technologies offer potential for monitoring and prevention, adoption rates remain low due to usability barriers and acceptance issues. **Objective**: This study explored how older adults perceive home-oriented smart technologies for fall prevention, aiming to identify specific design priorities that preserve autonomy while ensuring safety. **Methods**: Four semi-structured focus groups were conducted with 45 older adults (aged 71-91) residing in independent and senior community settings in Florida. Transcripts were analyzed using a hybrid deductive thematic analysis approach to categorize perceptions of risk, usability barriers, and design preferences. **Results**: Three primary themes emerged: (1) ***Attribution of Risk***, where participants often internalized falls as personal failures or "clumsiness," leading to a "pride factor" that discourages the use of visible help-seeking devices; (2) ***The Digital Burden***, where maintenance, false alarms, and subscription costs were viewed as "imposed taxes" rather than supports; and (3) ***Design Preferences***, revealing a strong preference for "invisible" or passive environmental sensors (e.g., automated lighting, radar) over active wearables. **Conclusions**: Current "active" fall detection technologies (wearables, buttons) conflict with older adults' desire to maintain dignity and minimize cognitive load. Future development should prioritize Passive Assistive Technology that requires zero user interaction to effectively bridge the gap between safety needs and user acceptance.

INTRODUCTION

Falls are one of the most common and serious health challenges affecting older adults living at home, contributing to injuries, hospitalizations, and increased mortality [1,2]. More than one in four adults aged 65 years and older experience a fall each year, and these events often result in injuries that limit mobility and participation in daily activities [2]. Beyond the immediate physical harm, falls can accelerate functional decline by restricting activities of daily living, increasing fear of movement, and reducing quality of life [3]. For many older adults, the ability to remain safely at home is closely tied to independence and overall well-being, yet age-related changes in balance, strength, reaction time, and sensory function make everyday mobility tasks such as walking and navigating stairs more difficult and hazardous [1,4].

Home-oriented smart technologies have been increasingly explored as a strategy to support safe aging-in-place by enhancing safety and maintaining autonomy through in-home monitoring and assistance [5,6]. These systems may include environmental sensors, wearable devices, voice assistants, and data-driven monitoring intended to reduce fall risk and support safer mobility in the home [6,7]. In principle, smart technologies can contribute to fall prevention by identifying hazards, detecting instability or abnormal movement patterns, and enabling rapid response following a fall event [7]. However, despite the promise of these technologies, adoption and long-term use remain limited, in part due to usability barriers, privacy concerns, cost, a lack of personalization, and low trust in automated monitoring and alerts [8,9]. If these systems are not designed around older adults' preferences, routines, and perceptions of independence, they may not be accepted or sustained even when they address important safety needs [8,9]. Current literature has identified these barriers broadly, but there remains limited qualitative insight into the specific reasons these challenges persist among older adults living independently, particularly regarding how they interpret and experience these technologies in everyday life.

The goal of this study was to explore how older adults perceive home-oriented smart technologies for fall prevention and to identify user priorities that can guide the development of effective, acceptable, and autonomy-preserving solutions. We conducted four focus groups with older adults to understand perceived fall-related risks in the home, desired smart technology features, barriers to acceptance, and expectations for how these systems should support independence. This research was guided by the following research questions: **1)** What are the specific usability barriers and design preferences of older adults regarding smart home technologies for fall

detection and prevention? **2)** How do older adults perceive the utility of technologies in maintaining independence versus the perceived threat to their autonomy?

METHODS

We conducted four semi-structured focus groups (N=45) with older adults residing in independent living and senior community settings in Gainesville, Florida. This study was determined to be exempt by the University of Florida Institutional Review Board (UFIRB Exempt: HSPET00047415). Participants ranged in age from 71 to 91. The sample included individuals with varying mobility levels, ranging from fully independent to those utilizing assistive devices (e.g., canes, walkers). Sessions were audio-recorded, transcribed, and analyzed using thematic analysis to identify specific usability barriers and design preferences regarding fall detection technologies. Focus groups were chosen over individual interviews to encourage discussion and uncover shared experiences related to technology use and safety. Audio recordings were transcribed using the Zoom cloud service. A thematic analysis approach was employed to identify barriers and facilitators related to fall detection and home safety technologies. Transcripts were coded to categorize user feedback into three primary domains: (1) perceptions of fall risk, (2) usability barriers of current devices, and (3) design preferences for future technologies. Data related to medication management, leisure activities, and other non-relevant domains were excluded to maintain focus on the scope of this manuscript.

Transcripts were analyzed using a hybrid deductive thematic analysis approach. Initial keyword extraction and pattern recognition were conducted using Large Language Model (LLM)–assisted tools to support the identification of high-frequency terms related to "falls," "usability," and "safety." All LLM-assisted outputs were subsequently subjected to manual verification. Two independent human coders reviewed and validated all extracted keywords and emergent patterns against the original verbatim transcripts, resolving discrepancies through discussion to ensure analytic rigor and credibility.

RESULTS

Participants were 45 older adults aged 71 and older, as shown in Table 1; the majority used technology regularly, including smartphones, tablets, and computers. The qualitative analysis showed that older adults viewed smart home technology as a solution to the main challenges of aging in place. Participants expressed their willingness and desire to integrate technology into their routines, both physical and social, without the cognitive burden.

Table 1: Demographics (Age, Gender, Assistive Device usage).

Category	Value	Percentage
Age	76-80	42.9%
	81-85	35.7%
	86+	14.3%
	71-75	7.1%
Living Situation	Live Alone	64.3%
	Live with Spouse/Partner	28.6%
	Assisted Living	7.1%
Type of Residence	Senior Living (Non-assisted)	78.6%
	Apartment/Condo	14.3%
	Senior Living (Assisted)	7.1%
Experience with Tech	Regularly use devices	78.6%
	Regularly/Occasionally use	14.3%
	Regularly use (with difficulty)	7.1%
General Health Status	Manage without assistance	78.6%
	Need some assistance	21.4%

The thematic analysis revealed a gap between the commercial availability of fall detection devices and the lived experience of older adults. Participants consistently described current technology as an "imposed" burden rather

than a supportive tool. Findings are categorized into three primary domains: Risk Attribution, Usability Barriers, and Design Preferences. Table 2 provides a matrix of these themes mapped to specific design requirements.

Table 2: Themes & Representative Quotes

Core Theme	Participant Voice (Quote)	Engineering/Design Implication
Risk: The "Pride" Factor	*"I kept saying I did not fall. I just sit down... The only thing that I heard was my pride."* (P2)	**Passive Detection:** Systems must detect falls via ambient sensors (radar/camera) without requiring the user to press a button or admit vulnerability.
Barrier: False Alarms	*"I was trying to dig in my garden... and the [watch] said, 'Have you fallen?'"* (P4)	**Context-Awareness:** Algorithms must distinguish between high-METs leisure activities (gardening) and actual falls to prevent device abandonment.
Barrier: Maintenance	*"I unplugged it, and now it can't even talk to me... I've got to re-hook it up again."* (P2)	**Zero-Configuration:** Devices must be "plug-and-play" with auto-reconnection capabilities after power loss. No complex pairing.
Preference: Automation	*"I want my microwave to be smart. It knows what to do. I don't need to do anything about it."* (P1)	**Autonomous Action:** Shift design focus from "Information" (apps/dashboards) to "Actuation" (robotic assistance, auto-lighting).

Theme 1: Attribution of Risk - The Tension Between Physiology and Environment

For participants, fall risk was a twofold failure of the internal "machine" and external "danger zones". However, the active use of help-seeking devices is hindered by psychological barriers, such as pride.

1.1 Internal Disconnect: A delay is sensed between what the brain intends and the body executes. As one participant put it, "My brain does not... The brain and the body do not connect" (File 1), and another used a mechanical metaphor: "I had a good engine, but the electrical system isn't so great" (File 4).

1.2 The "Pride" Factor: Admitting a fall is stigmatized, leading users to reframe incidents. "I kept saying I did not fall. I just sat down. The only thing that I heard was my pride and my butt" (File 2). This suggests that manual buttons, for example, PERS, are often ignored to preserve dignity.

1.3 Environmental Awareness: Bathrooms were universally recognized as transitional areas of vulnerability. Participants reported a conflict between the dangers of the environment and their own attention: "If I look at my feet, I'm gonna fall. keep my head up and look out" (File 1).

Theme 2: The Digital Burden – Why Current Tech Fails

There exists a "capability gap" in understanding technology as an imposed tax versus an approach -enhancing.

2.1 Imposed Complexity: Even technologically savvy seniors face the exhaustion of having to keep adapting. “We all adapted to technology the hard way. It was, in essence, imposed ... We were always catching up” (File 1).

2.2 Maintenance & Financial Fatigue: The adoption of boundless SaaS (Software as a Service) and incessant troubleshooting has induced abandonment. One user complained, "The programs became subscriptions. I can no longer afford that monthly" (File 2). For older adults on fixed incomes, the shift from "buying a product" to "renting a service" is a major financial barrier to safety. Physical Maintenance is another inhibitor: "I unplugged it, my smart speaker. Now I've got to re-hook it up again" (File 2).

2.3 Reliability and False Alarms: A "cry wolf" effect is a problem present with wearables; "The [watch] said, 'Have you fallen?' I said, yeah, but no! Turned off the 911" (File 4).

Theme 3: Design Preferences – The Desire for "Invisible" Support

There was an evident shift from information-provisioning apps to task execution via automation.

3.1 Automation instead of Information: There is a desire to have robots take care of the work involved in daily life. "I want my microwave to be smart. It knows what to do. I don’t need to do anything about it" (File 1). Someone also wanted a robot for "washing my clothes. and putting them back" (File 2).

3.2 Passive Environmental Safety: This category featured designs that were highly valued if they involved zero user input or action. "The infrasound, activated lights, so when you get up to go to the bathroom, that's when the light goes on" (File 3).

DISCUSSION & DESIGN IMPLICATION

Our findings reveal a critical psychological disconnect between the design of current fall-detection devices and older adults' self-perceptions. While the industry prioritizes active wearables, our participants frequently attribute falls to internal "clumsiness" or a disconnect between the "brain and body" rather than to external hazards. This internalization of risk creates a "Pride Factor," where users are reluctant to engage with devices that serve as constant, visible reminders of their frailty. This aligns with previous research suggesting that if systems are not designed around older adults' perceptions of independence, they will be rejected even if they address safety needs [8, 9]. Consequently, manual help-seeking buttons or conspicuous wearables are often ignored or removed to preserve dignity.

Despite the promise of smart home technology to support aging in place [5], our participants described these tools as an "imposed" burden rather than a support system. The requirement for constant maintenance, such as charging batteries, managing false alarms, and navigating subscription models, acts as a functional tax on the user. For example, the need to "re-hook" devices after power failures or manage recurring costs led to device abandonment among our group. This aligns with the existing literature, which identifies usability and cost as the primary barriers to the long-term adoption of in-home monitoring systems [6, 8]. When technology demands high cognitive or financial maintenance, it paradoxically reduces the independence it is meant to preserve.

To bridge the gap between safety requirements and user acceptance, design philosophy must shift from "active engagement" to "passive support". Participants expressed a clear preference for automation that requires zero user input, such as infrasound-activated lighting or "rubberized" bathroom environments, effectively requesting a "set and forget" approach. Future development should focus on Invisible/Passive Assistive Technology (IPAT). By utilizing environmental sensors (e.g., radar, ambient optical sensors) [7] rather than on-body wearables, engineers can provide fall detection that respects the user's desire for autonomy and eliminates the "digital burden" of daily maintenance [5, 6].

CONCLUSION

Falls remain a leading cause of functional decline and mortality among older adults. While the commercial market has responded with an array of wearable monitors and subscription-based alert systems, this study highlights a significant disconnect between these solutions and the lived experience of independent seniors. We found that the "digital burden" of maintenance, combined with the psychological stigma of visible assistive devices, frequently leads to non-compliance and device abandonment. The future of effective fall prevention does not lie in smarter wearables, but in smarter environments. To ensure widespread adoption and sustained use, the field must pivot toward passive integration, technologies embedded in home infrastructure that require no user interaction and operate invisibly to protect older adults without challenging their sense of dignity or independence.

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